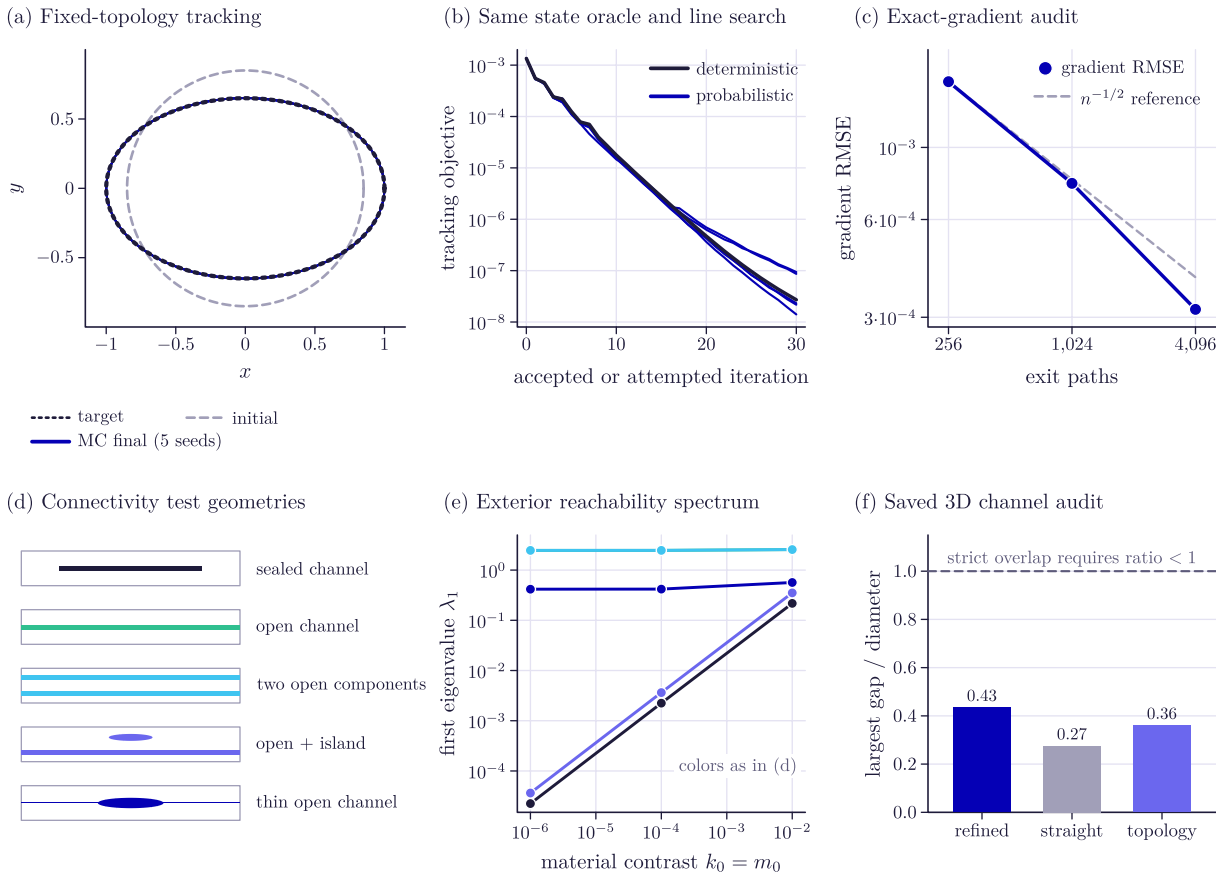

(a) Fixed-topology tracking
0.5
0
−0.5
y
−1
−0.5
0
0.5
1
x
target
initial
MC final (5 seeds)
(b) Same state oracle and line search
deterministic
probabilistic
tracking objective
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}
10^{-8}
0
10
20
30
accepted or attempted iteration
(c) Exact-gradient audit
gradient RMSE
$n^{-1/2}$ reference
gradient RMSE
10^{-3}
$6{\cdot}10^{-4}$
$3{\cdot}10^{-4}$
256
1,024
4,096
exit paths
(d) Connectivity test geometries
sealed channel
open channel
two open components
open + island
thin open channel
(e) Exterior reachability spectrum
first eigenvalue λ_1
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
colors as in (d)
10^{-6}
10^{-5}
10^{-4}
10^{-3}
10^{-2}
material contrast $k_0 = m_0$
(f) Saved 3D channel audit
strict overlap requires ratio < 1
largest gap / diameter
1.0
0.8
0.6
0.4
0.2
0.0
0.43
0.27
0.36
refined
straight
topology